UNIT -V

Q.5 (a) Solve the equation $x^4 - 2x^3 - 21x^2 + 22x + 40 = 0$ whose roots are in A.P. **06**

(b) Solve by Cardan's method $x^3 - 9x^2 - 9x - 15 = 0$. **06**

(c) Verify the De-Morgan's laws for the fuzzy sets given by **06**

$X = \{1, \quad 2, \quad 3, \quad 4, \quad 5, \quad 6, \quad 7, \quad 8, \quad 9, \quad 10\}$

$A = \{(2,0.1),(3,0.3),(4,0.5),(5,1),(6,0.2),(7,0.4),(8,0.6),(9,0.8)\}$

$B = \{(3,0.2),(4,0.4),(5,0.6),(6,0.8),(7,1)\}$.

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Roll No.............

2023023

BE-I EXAMINATION FEB.-MARCH'2023
ETC (A, B) BRANCHES (RE)
2AMRC1: Applied Mathematics-II

Duration: 3Hrs. **Max.Marks:60**

Note: Attempt any **two** parts from every question. Questions should be solved at **one place.** All questions carry equal marks. Any assumption made answering the questions should be stated. Assume suitable data whenever necessary.

UNIT –I

Q.1

(a) If $A = \begin{bmatrix} 3 & -3 & 4 \\ 2 & -3 & 4 \\ 0 & -1 & 1 \end{bmatrix}$, find two non-singular matrices P and Q such that $PAQ = I$. Hence find A^{-1}. **06**

(b) Test for consistency and solve **06**

$$x_1 - x_2 + x_3 - x_4 + x_5 = 1;\ 2x_1 - x_2 + 3x_3 + 4x_5 = 2;$$
$$3x_1 - 2x_2 + 2x_3 + x_4 + x_5 = 1;\ x_1 + x_3 + 2x_4 + x_5 = 0.$$

(c) Find the eigen values and eigen vectors of $A = \begin{bmatrix} 2 & -2 & 2 \\ 1 & 1 & 1 \\ 1 & 3 & -1 \end{bmatrix}$. **06**

UNIT -II

Q.2 (a) Find the differential equation whose set of independent solution is $[e^x, xe^x, x^2e^x]$. **06**

(b) Solve $\frac{d^2y}{dx^2} - y = x\sin x + e^x + x^2e^x$. **06**

(c) Solve the simultaneous equations: $\frac{dx}{dt} + 4x + 3y = t,\ \frac{dy}{dt} + 2x + 5y = e^t$. **06**

UNIT -III

Q.3 (a) Form Partial Differential Equation from **06**

(i) $z = yf(x) + xg(y)$. (ii) $2z = \frac{x^2}{a^2} + \frac{y^2}{b^2}$.

(b) Solve $(x^2 + y^2)p + 2xyq = (x + y)z$. **06**

(c) Solve $\frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial x \partial y} - 6\frac{\partial^2 z}{\partial y^2} = x^2 \sin(x + y)$. **06**

UNIT -IV

Q.4 (a) A and B take turns in throwing two dice, the first to throw 10 being awarded the prize. Show that if A has the first throw, their chance of winning are in the ratio 12:11. **06**

(b) Define Binomial distribution also find (i) mean and (ii) variance of B.D. **06**

(c) Calculate the coefficient of correlation between the marks obtained by 8 students in mathematics and statistics: **06**

Students	A	B	C	D	E	F	G	H
Mathematics	25	30	32	35	37	40	42	45
Statistics	08	10	15	17	20	23	24	25